Jaiden Peltonen

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A diligent doctoral candidate with a love for teaching and research. My research is focused on star formation in nearby galaxies combing the latest observations with James Webb Space Telescope with cutting edge modelling techniques.

Keywords: Protostars; Interstellar clouds; Local Group

Education

(2023-2027) PhD | PHYSICS | UNIVERSITY OF ALBERTA

- Supervisor: Erik Rosolowsky

(2021-2023) MSc | PHYSICS | UNIVERSITY OF ALBERTA

- Supervisor: Erik Rosolowsky

(2017-2021) BSc - HONOURS -FIRST CLASS | UNIVERSITY OF CALGARY

- Major: Astrophysics

Publications

Peltonen, J., Rosolowsky, E., Ginsburg, A., et al. *Introducing PxP: A Population Synthesis Framework for Predicting Young Stellar Object Properties* (2026) The Astrophysical Journal, 1001, 170

Peltonen, J., Rosolowsky, E., Williams, T. G., et al. *JWST reveals star formation across a spiral arm in M33* (2024) Monthly Notices of the Royal Astronomical Society, 527, 10668

Peltonen, J., Rosolowsky, E., Johnson, L. C., et al. *Clusters, clouds, and correlations: relating young clusters to giant molecular clouds in m33 and m31* (2023) Monthly Notices of the Royal Astronomical Society, 522, 6137

Rodríguez, M. J., Indebetouw, R., …. **Peltonen J.**, et al. *Dust-embedded Star Formation: Bridging Magellanic Cloud Studies of Massive Young Stellar Objects to Nearby Spiral Galaxies* (2026) The Astrophysical Journal, 1006, 144

Román-Zúñiga, C., Palau, A., … **Peltonen, J.**, et al. *Comparing the Mgas—Nyso relation inside a giant molecular cloud* (2026) Monthly Notices of the Royal Astronomical Society, 547, 510

Wainer, T., Dalcanton J., … **Peltonen, J.**, et al. *The Timescales of Embedded Star Formation as Observed in STARFORGE* (2026) The Astrophysical Journal, 998, 215

Sarbadhicary, S., Roslowsky, E., … **Peltonen, J.**, et al. *A First Look at Spatially Resolved Infrared Supernova Remnants in M33 with JWST* (2025) The Astrophysical Journal, 989, 138

Richarson, T., Ginsburg, A. … **Peltonen, J.**, et al. *A Framework for Modeling the Evolution of Young Stellar Objects* (2025) The Astrophysical Journal, 989, 95

JWST Proposals

ACCEPTED CO-I

GO 3436: Unmixing the ISM: Identifying Dominant Physical Effects with JWST/MIRI Mapping of M33

GO 9622: A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

CURRENT CYCLE PI

ASOOLA: Andromeda Survey of Obscured-objects Over Lifetimes of Accretion

Ice and Evolution of YSOs - MICASA: M33 Icy-YSO Chemistry And Spectral Analysis

Conferences

ORAL PRESENTATIONS

- Star Formation Fields (Crete IV), Greece (2026)
- The Olympian Symposium, Greece (2026)
- Star Formation in Different Environments, Vietnam (2025)
- Star Formation Workshop McMaster (2024)
- CASCA AGM (2023)
- Clusters at McMaster (2022)
- The 6th Annual UCalgary PHAS Symposium (2021)
- Undergraduate Research in Science Conference of Alberta (2021)

POSTER PRESENTAION

- The First Year of JWST Science Conference, USA (2023)
- CASCA AGM (2021)

Awards

- Physics Graduate Student Travel Award 2026
- Alberta Graduate Excellence Scholarship 2023
- 1st Place at URSCA 2021
- Imperial Oil Limited Award 2016, 2017, 2018, and 2019

Workplace and Teaching Experience

GRADUATE TEACHING AND LEARNING: PRACTICUM | UNIVERSITY OF ALBERTA | SEPT 2024

LAB TEACHING ASSISTANT | UNIVERSITY OF ALBERTA | 2022-2026

OBSERVATORY | UNIVERSITY OF ALBERTA | 2021-2026

- Ran observatory tours and presentations for school groups

GRADUATE TEACHING AND LEARNING: FOUNDATIONS | UNIVERSITY OF ALBERTA | 2021

LEAD LAB TEACHING ASSISTANT | UNIVERSITY OF CALGARY | 2020

- Lead the team of teaching assistants for Mechanics labs

NUCLEAR EDUCATOR | UNIVERSITY OF CALGARY | 2020

- Created educational videos and 3D models about nuclear energy

LEARNING ASSISTANT | UNIVERSITY OF CALGARY | 2020

- Assisted students in Electromagnetism II

LAB DEVELOPMENT | UNIVERSITY OF CALGARY | 2019

- Developed and tested first and second year physics labs

Science Volunteering

- CASCA GSC UAlberta representative 2024-2025
- Hosted the weekly seminar and journal club for the astrophysics group at the UAlberta 2023-2024
- Host at URSCA 2019
- Volunteered at the Rothney Astrophysical Observatory doing science education to the public 2018-2019